



Seminar Paper

Analysis of Public Formula 1 Data

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Submitted on: 15th August 2023



Abstract

This project is divided into several stages, from data collection to the development, visualisation and evaluation of the predictive model using the machine learning formula. The Ergast API and Formula 1 telemetry data from the 2018-2023 seasons are the primary data sources utilised in this work. The methodology aims to predict the fastest lap times in the qualifiers for the 2023 season.

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1 Introduction

Formula 1 (F1) is the most expensive motorsport that produce the millions of data through combining speed, strategy, car, tracks and skill in a spectacle that captivates fans around the world. Each race, known as a Grand Prix, is a battle not just between drivers but also between the teams that engineer the cars. Behind the glamour and excitement of each race lies a world of intense competition, where fractions of a second can make the difference between victory and defeat.

In this rapidly evolving environment, data plays a critical role. Teams harness vast amounts of information—from car telemetry, driver performance, weather conditions, to tire strategies—to gain a competitive edge. For example, between 1950 and 1957, events ran for more than 300 km (190 mi) or three hours. From 1973 to 1980, races had to last either 321.87 km (200.00 mi) or two hours, whichever came first. Distances of between 250 and 320 km (160 and 200 mi) or two hours were used from 1981 to 1984. As of the 2023 Belgian Grand Prix, 1,091 World Championship events have been held over 73 seasons in 34 countries and under 53 race titles at 76 racing circuits.

As the Formula 1 happens almost major continents of the world that requires accurate weather precision and track data. These data provides a rich source of material for predictive analysis. With the right approach, it is possible to leverage this data to forecast future race outcomes, offering insights that can be valuable not only to the teams themselves but also to fans and stakeholders in the sport.

The objective of this report is to build a predictive model that forecasts the outcomes of the 2023 Formula 1 season based on historical data from the 2021 and 2022 seasons. This involves an in-depth analysis of various features such as driver performance, team strategy, car specifications, and external factors like weather conditions. By examining these features, we aim to understand the patterns and trends that lead to success in Formula 1 and to make informed predictions about future races.

In this report, we will Conduct exploratory data analysis (EDA) to understand the underlying structure and trends in the data. Investigate the temporal trends of driver and team performances across the 2021 and 2022 seasons. Analyse the correlation between different features and race outcomes. Develop and validate predictive models using machine learning techniques. Use the best-performing model to predict the outcomes of the 2023 Formula 1 season. Conduct scenario analysis to understand how different factors might impact the 2023 season. Significance of the Study Predictive analysis in Formula 1 is not just an academic exercise; it has real-world applications. Teams can use predictive insights to refine their strategies and optimise performance. Fans can gain a deeper understanding of the sport they love, and broadcasters and sponsors can use predictive data to enhance their coverage and engagement with the audience. Furthermore, the methods and insights derived from this analysis may have applications beyond Formula 1, offering lessons for data-driven decision-making in other complex, competitive environments.

2 Data Summary

2.1 Overview

As already mentioned the main objective of this project work is to be able to predict the results of the Qualifiers of the season 2023, which have been run this far. The available data is divided into seasons ranging from 2018 to 2023 each season contains about 22 weekend where the races take place. Further are those weekends split into sessions. A typical race weekend consists of the following sessions:

- Three practice runs (FP1, FP2, FP3)
- One Qualifier run (Q)
- One actual Race (R)

The practice runs stand the purpose so that the drivers can test their cars and equipment for errors and can adjust it without any impact on the final results. The Qualifier run then determines the starting position of the drivers on the final race. Here the fastest driven lap decides who will start in the so called pole position.

To be able to predict the results of the Qualifier the data of the three practice runs and the races of the past were extracted. The practice runs usually consist of about 15 laps each that are driven on the track. The actual race then proceeds to be about 50 to 60 laps of data depending on the driven Grand Prix. So for each weekend the collected data for each driver consists of about 100 laps of data.

During those events multiple terabytes of data is being collected in order to analyse and further improve the drivers and cars performance. As most of the data is kept private so that other teams can not spy on their progress only a limited amount of data is publicly available. The unprocessed data consists of 40 columns describing important conditions of the race. A rough subdivision of the columns can be made by the following categories:

- timing data
- speed data
- driver information
- time and day
- tire information
- track information
- weather information

As the Qualifier is decided by the fastest lap time of the drivers this aspect is therefore the most important target of this task.

The data extraction for this project was performed using the R package Fastf1R created by Casanova and Bulsink (2022) which uses the FastF1 python packages data stream created by Schaefer (2019) and the Eargast API (<http://ergast.com/mrd/>) to access the data.

2.2 Data preparation

The extracted data spans the years 2021, 2022, and 2023, totalling approximately 100,000 rows. All this data was consolidated into a single dataset to streamline further preparation, visualisation, and machine learning processes.

To clearly distinguish between each collected session, two new columns were introduced: one indicating the year in which the session took place and the other identifying the specific Grand Prix or track where the sessions were conducted. As the primary focus of this dataset is lap times, the subsequent preparations largely revolved around these figures.

Upon extraction from the Ergast API, the dataset exhibited several imperfections. In some instances, specific pieces of data were missing; in more severe cases, entire sessions for certain drivers were absent. Errors were also encountered during the data collection phase for some sessions, resulting in a complete lack of data for certain practice runs. In some instances, the absence of this data could be attributed to practice runs being cancelled during race weekends due to external issues.

The initial step in the data preparation involved removing all rows lacking lap time information for the sessions. Subsequently, the weather data, which was extracted in list form, needed to be restructured to facilitate seamless processing in later stages of the project.

2.2.1 Outlier detection

With all remaining data presented in numeric, logical, or character format, the process of outlier detection was undertaken. This detection was guided by a boxplot analysis of the data.

As visible in the figure 2.1 there are many outliers present in the dataset when observing the lap times. These outliers can occur due to different reasons, in the event of a crash of one of the drivers during a race the safety car takes the lead while all drivers have to slow down significantly to avoid further crashes. The safety car is also introduced in the case of bad weather conditions which can lead to much higher crash possibilities.

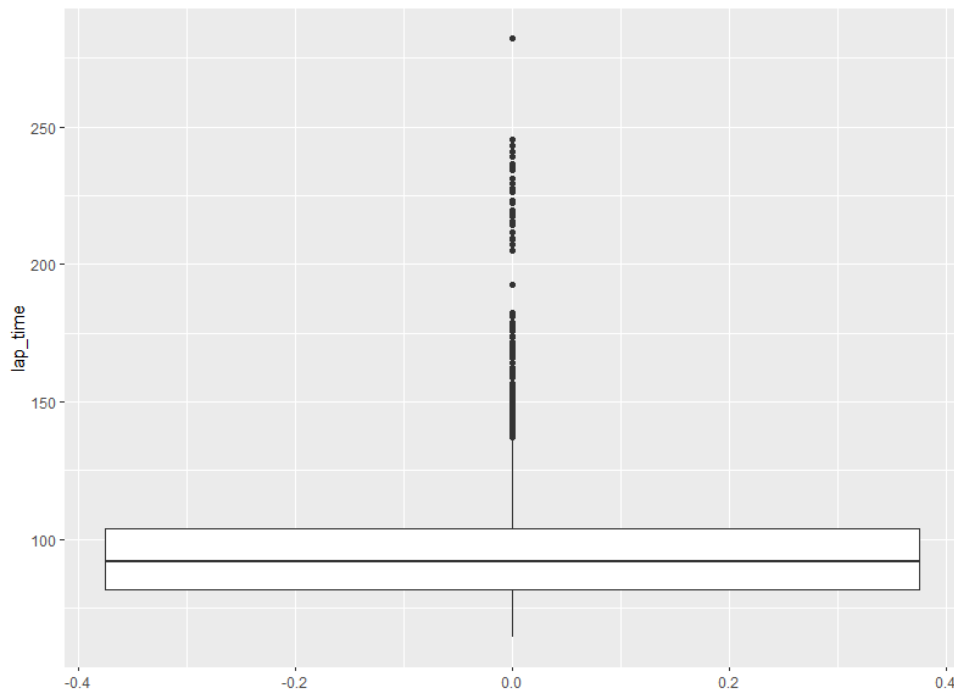


Figure 2.1: Boxplot of unprocessed data

Further reasons are that the drivers might not push their car to the absolute limit in their first rounds of the practice runs and therefore just drive way slower in order to get a feeling for the car.

The structure of a boxplot shows the mean of the data and the interquartile range (iqr) in white. The boxplot usually has 2 bounds where data points count as outliers. This is done by calculating the upper and the lower quantile of the data. Then to determine the outer bound the iqr is multiplied by 3 and then added to the upper quantile and subtracted to the lower quantile.

When applying the outer bounds only 36 rows of data were eliminated, which only contains the largest outliers of the dataset. However as the dataset contains more than 100,000 rows of data the lower bound can be used to further eliminate outliers. The lower bound adds and subtracts only the values of 1.5 times the iqr. When applying the lower bounds more than 2500 rows of data were eliminated from the dataset.

When looking further into the selection it is almost obvious that the lower bounds are always useless as outliers are always result in higher lap times as it is a time related variable. One could therefore argue to further lower the upper bound to eliminate outliers on shorter tracks which are not detected for now. As each track of the data is not equally in length the outlier detection has to be calculated carefully. When lowering the bounds even further tracks like the Canadian with a fastest lap time of about 108 seconds could find themselves in a reduction of data which does not count as an outlier. A finer tuning could be done by calculating the bounds for each track of the dataset and then eliminating outliers depending on the track.

As this would take up way more computing, we found that the elimination of 2500 rows or 2% of the dataset was sufficient enough. As there are no tracks which are significantly shorter than the usual tracks in the dataset a specific outlier detection for just one track was also not necessary. As visible in figure 2.2 the data does now contains way less outliers than before. With some outliers still being present the data does now not contain large outliers which were could be affecting the results by a larger margin.

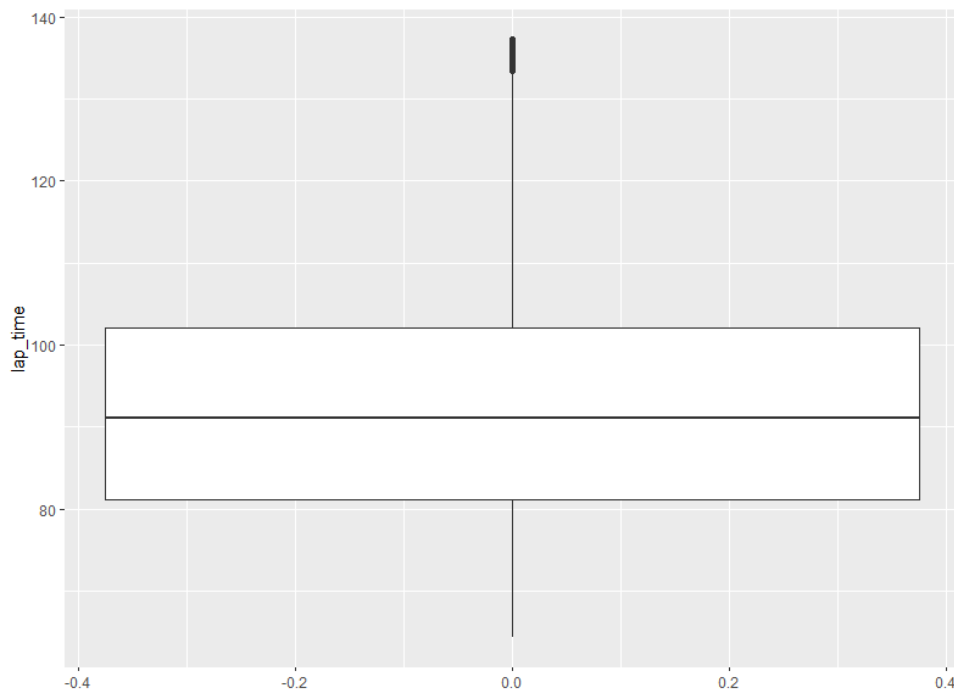


Figure 2.2: Boxplot of processed data

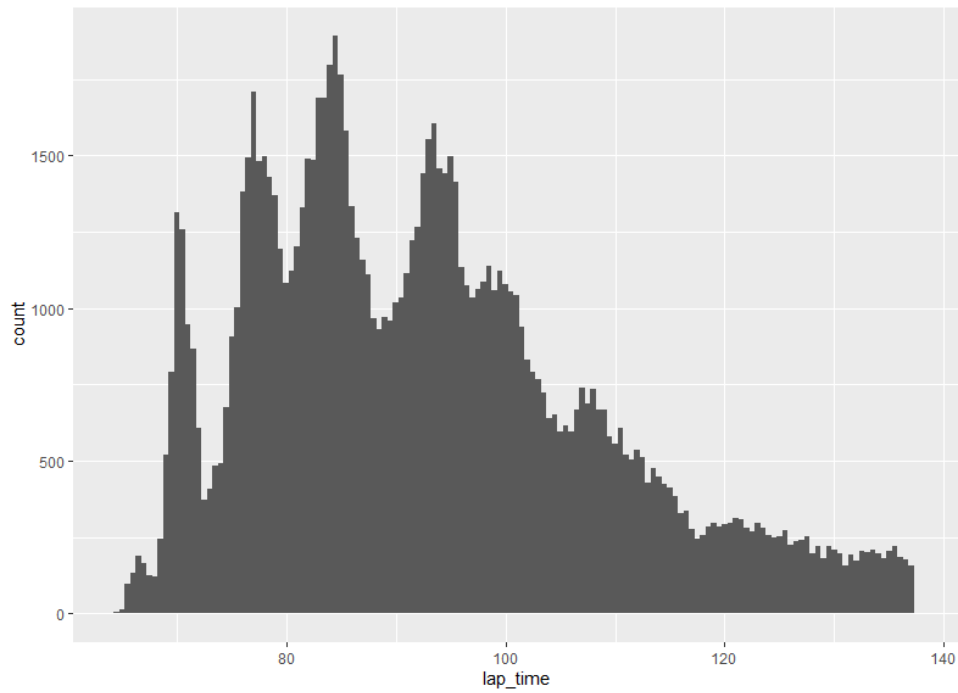


Figure 2.3: Histogram of processed data

A look into the histogram of the processed data in figure 2.3 shows large spikes around the 85 and 95 second mark while still containing a significant amount of data upwards of 120 seconds. The data does also indicate a large spike around the 70 second mark these datapoints mark the laps of the Austrian GP which is present four different times within this dataset and therefore states the most data.

2.2.2 First feature selection

In further steps the data is now being reduced by insignificant features. The pre-selection of the features is done carefully as a reduction in features can significantly reduce the effectiveness of the machine learning model.

The eliminated features were columns containing the time of day and the day and month when the race took place. Further more were the amount of laps, stints and cars ahead eliminated as all of these factors differ greatly when comparing practice data to race data. Other eliminated columns showed information if the data was retrieved accurately and if some parts were deleted. Those factors do also not represent any important data of the actual race.

After the trimming of features a total of 23 columns were left in the dataset. The feature now contained the following information:

- Sector timings and speed
- Driver name and number
- Tyre life, compound and freshness
- Air and track temperature
- wind speed and direction
- Air pressure and humidity
- Year and Grand Prix driven

2.2.3 Further pre-processing

The next step processed the missing datapoints in the sector speed and timing data. Throughout the data about 25,000 rows had missing speed data and another 1000 were missing timing data. These missing datapoints have to be dealt within the pre-processing as machine learning algorithms like the regression tree learner can not deal with missing data points.

One approach would be to just eliminated the rows with missing datapoints from the dataset. As this procedure would subsequently get rid of almost 25% percent of the collected data another approach was taken. The new process would include filling the missing datapoints with the mean values of the rest of the data. This approach does lead to a distortion of the speed data of the dataset. But with just one of the four columns regarding speed data being adjusted the distortion would be held to a minimum as the other three datapoints were not affected.

In the next step all character based columns were reformed into numerical form. This transformation did affect the columns regarding tyre compound and the name of the Grand Prix. During the transformation about 1500 rows of data with missing tyre compounds were eliminated from the dataset. This transformation had to be done to be able to perform a benchmark of learners in the later stages. The k-nearest neighbour learner can for example not deal with data of the type character.

Another important task of the data preparation is to set up the data sets on which the predictions are being done. This task follows the same procedure as the training data, including the unlisting of the weather data, the adding of the column describing the year and removing any rows with missing lap time data. Now the decision stands for which driver the prediction should be calculated. Looking into the current driver standing of the season 2023, one could see that the top 4 drivers are currently:

- Max Verstappen (VER)
- Sergio Perez (PER)
- Fernando Alonso (ALO)
- Lewis Hamilton (HAM)

As these four drivers are ahead of their other colleagues regarding their standings, the prediction will be made for these drivers (Formula One 2023). Regarding the data preparation 4 different dataset were created with the best lap times of the qualifier for each Grand Prix summing up to 48 predictions in total.

3 Data Visualisation

3.1 Overview

Visualisation serves as a crucial tool for unravelling intricate datasets in the context of Formula 1 races. To understand the datasets we have followed the ergast API, fastf1 and formula1 website . In a sport defined by precision and the multitude of parameters shaping race outcomes, comprehending these data sets stands paramount. We delve into tailored visualisation techniques for a Formula 1 machine learning initiative spanning 2021 to 2023. The complexity of Formula 1, encompassing drivers, cars, tracks, sessions and strategies, yields extensive data sets covering lap times, telemetry, weather, tire performance, and beyond. We tried effectively to process and visualise, these data points proffer insights into race strategy, car performance, and predictive analysis. Before exploring visualisation techniques, critical data points are established, including real-time telemetry encompassing speed, throttle, brake usage, and more, lap times, weather conditions, tire behaviour, and driver performance metrics. Subsequently, the chapter delves into diverse visualisation approaches. Through the time series analysis elucidates performance trends across multiple years, that reveal patterns and anomalies. We also tried to understand the Lap time evolution graphs to expose performance fluctuations, while overlaying weather data onto performance metrics illustrates the impact of changing conditions. Heat maps, representing telemetry at circuit sections, shed light on driver behaviour and physical demands. Comparative visualisations facilitate performance assessment across drivers and teams using bar and spider charts. Predictive visualisation tools, integral to machine learning predictions, are outlined, including probability distribution plots. Given the global nature of races, geospatial visualisation aids in location-based insights. Circuit analysis through satellite data coupled with performance metrics provides insight into circuit-specific dynamics.

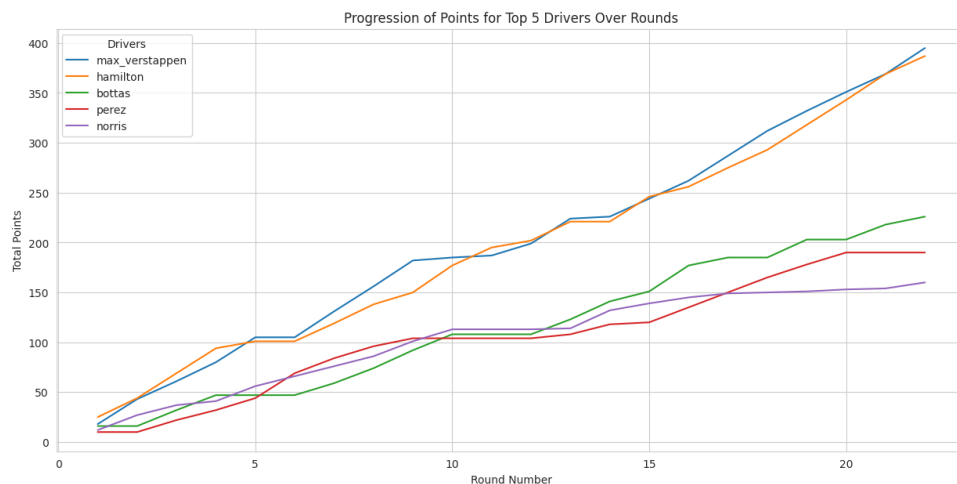


Figure 3.1: Progression of top 05 qualifiers of 2021

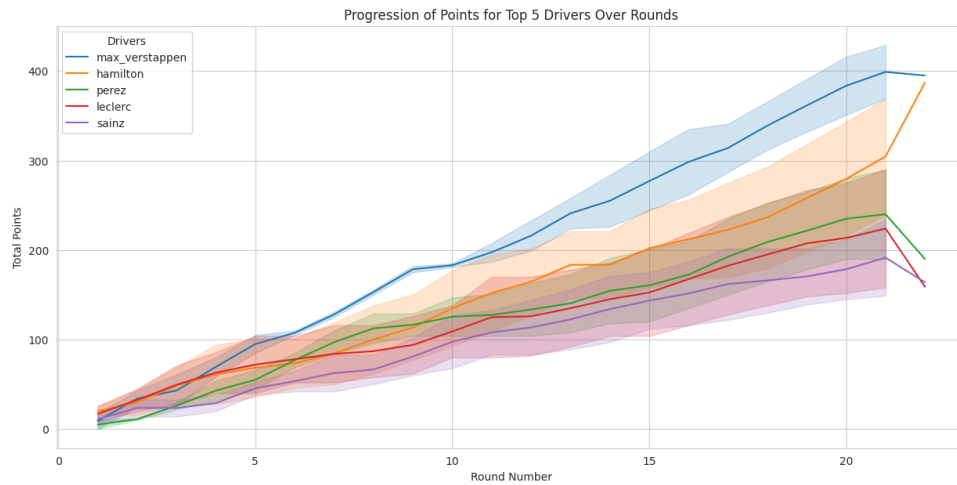


Figure 3.2: Progression of top 05 qualifiers of 2022

Empowering these visualisations are libraries such as Matplotlib, Seaborn, and Plotly, enabling static, animated, and interactive visual representations within the Python programming ecosystem.

At the very initial we tried to understand the last five years data. Max Verstappen from Red Bull, the leading driver exhibited a remarkable trajectory over the 2021 & 2022 datasets and the pits, consistently elevating its performance. In 2021 we see a smooth linear progression of his points. Hamilton from Mercedes, was close on his neck into the graphs. In 2021, Hamilton showcased a consistent performance, securing multiple podium finishes and victories. The graph illustrates his competitiveness in terms of average points per race, podiums, and wins, with Hamilton generally having an edge over Verstappen in these categories. However, Verstappen's strong presence is evident in his higher number of poles and fastest laps. The battle between the two drivers was fierce, with Hamilton ultimately clinching the World Championship title by a narrow margin.

Moving to 2022, the graph indicates a continuation of the competitive rivalry between Hamilton and Verstappen. Both drivers remained strong contenders, with Hamilton consistently earning points, podiums, and wins, while Verstappen continued to dominate in the categories of poles and fastest laps. The race for the championship remained tight, with Hamilton and Verstappen exchanging the lead throughout the season. The graph underscores the intense competition between the two drivers, as they pushed each other to their limits in pursuit of the championship title.

Overall, the provided graphs offer a comprehensive overview of Lewis Hamilton's performance over the two years, highlighting his battles with Max Verstappen and the dynamic nature of their rivalry on the Formula 1 stage.

3.2 Tire, Pit Stops

Formula 1 race results involves considering a multitude of strategies, and both tire data and gearshift strategy play significant roles in shaping race outcomes Here we choose the 2021 Max Verstappen Abu Dhabi race tire compound along with visualising the Lap Time vs Lap Number and therefore tried to understand impact of Tire management is crucial in Formula 1 due to the varying conditions of the track, weather, and race strategy. His team selected the soft tire initial first 12 race and after last pit Stops. Between these laps he most used the hard tires due to the characteristics, including grip level and durability. Predicting race results requires analysing how teams and drivers manage tire wear and degradation throughout the

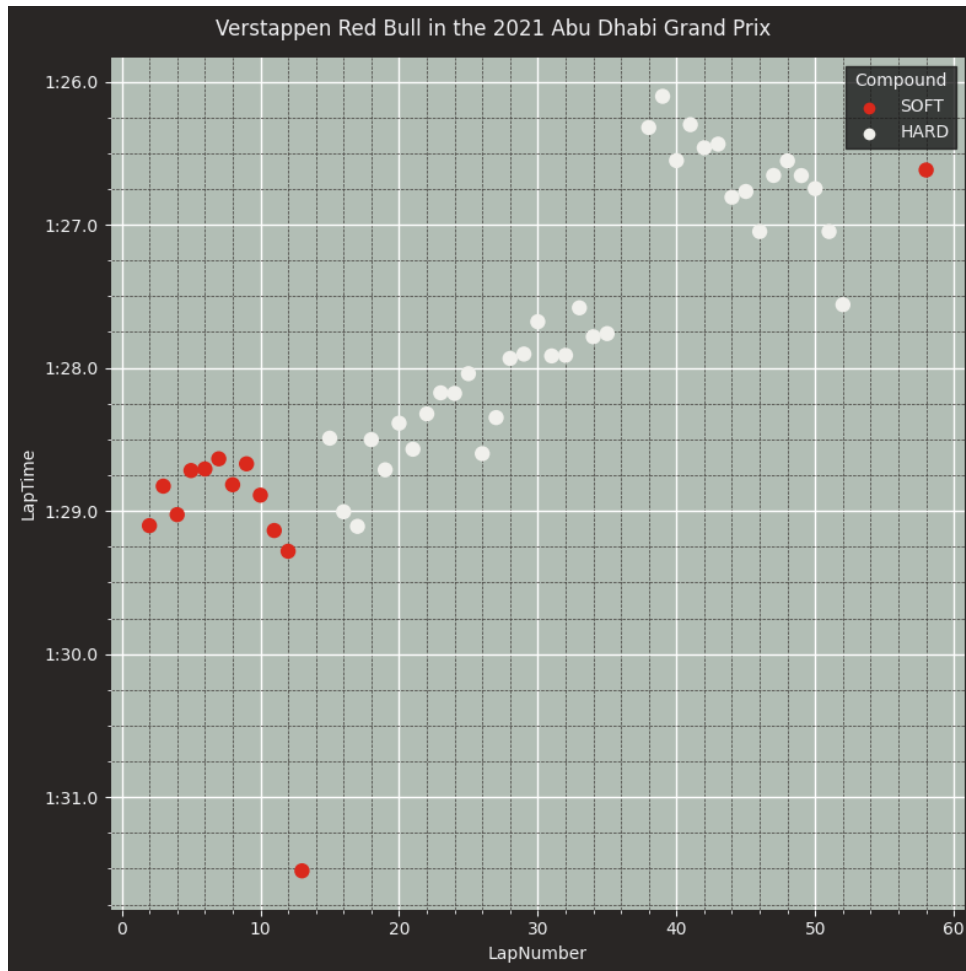


Figure 3.3: Max Verstappen in 2021 Abu Dhabi Grand Prix

race. **1. Tire, Pit Stops and Strategy:** Verstappen Team must decided to make pit stops for fresh tires. This decision depends on factors like tire degradation rate, track conditions, and race strategy as Abu Dhabi Grand Prix temperature 2021 Conditions: Mainly sunny with a few high clouds; Maximum temperature expected: 26 Celsius; Minimum temperature expected 20 Celsius; Chance of rain: 0Therefore, he choose to go for only most soft and Hard tires. **2. Stint Length:** Analysing historical tire data helps determine the optimal stint length for each tire compound. Longer stints can provide a strategic advantage by reducing the number of pit stops, but the risk of significant tire degradation must be managed. **3. Weather Conditions:** Tire choice can be influenced by weather conditions. 0 percent chance of Rain in Abu Dhabi and high humidity temperature changes significantly impacted tire performance. Predicting changing weather patterns and their effects on tire grip and wear is essential.

3.3 Gear Shift Strategies

Gearshift strategy affects both acceleration and fuel efficiency. All Drivers and teams makes calculated decision to optimise speed and conserve fuel over the course of the race. During the race, Max Verstappen was again locked in a battle with Lewis Hamilton for the championship title. A late Safety Car period presented Verstappen with an opportunity to pit for fresh tires. Red Bull decided to bring him in, and he switched to a set of soft tires. This strategic move allowed Verstappen to make a late charge in the final laps of the race.

Verstappen's gearshift strategy became essential as he aimed to close the gap to Hamilton,

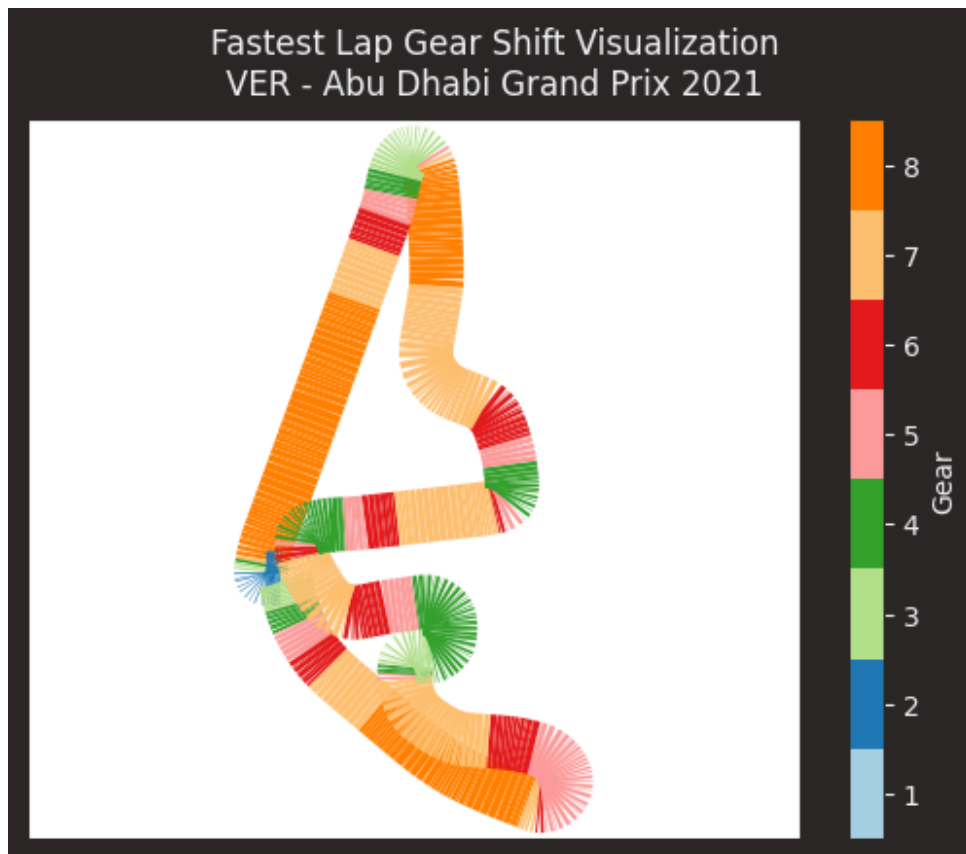


Figure 3.4: Max Verstappen 2021 Abu Dhabi Gear shifts

who was leading the race. The soft tires gave him better grip and performance, enabling him to push harder and close the time gap more rapidly.

With the race resuming for a single racing lap after the Safety Car period, Verstappen's fresher tires played a pivotal role in helping him gain a significant advantage over Hamilton. The difference in grip and performance allowed Verstappen to overtake Hamilton on the last lap and cross the finish line ahead of him, securing the race win and the World Championship title.

Throughout the race, Verstappen's gearshift strategies were crucial in managing his car's performance and extracting the maximum pace from his Red Bull. The decision to pit for fresh soft tires during the Safety Car period proved to be a masterstroke, allowing him to make the critical overtaking move on Hamilton.

In summary, Max Verstappen's gearshift strategies in the 2021 Abu Dhabi Grand Prix were integral to his victory and World Championship title. The decision to pit for fresh soft tires during a late Safety Car period enabled him to execute a thrilling overtaking maneuver on Lewis Hamilton and clinch the championship in a dramatic finale.

3.4 Weather Data Visualisation

The visualisation of the weather data add a interesting impact of the weather differences that the teams and driver have to deal with. The following figures show the variances of the weather circumstances during the qualifiers of the season 2023. The short tags on the x-axis indicate the driven Grand Prix. As most of the tags should be interpreted easily the following list does list the short tags as full names:

- AUS - Australian Grand Prix
- AZE - Grand Prix of Azerbaijan
- BAR - Grand Prix of Bahrain
- BEL - Belgian Grand Prix
- CAN - Canadian Grand Prix
- GBR - Grand Prix of Great Britain
- HUN - Grand Prix of Hungary
- MIA - Grand Prix of Miami (USA)
- MON - Grand Prix of Monaco
- OES - Austrian Grand Prix
- SAU - Grand Prix of Saudi Arabia
- SPA - Spanish Grand Prix

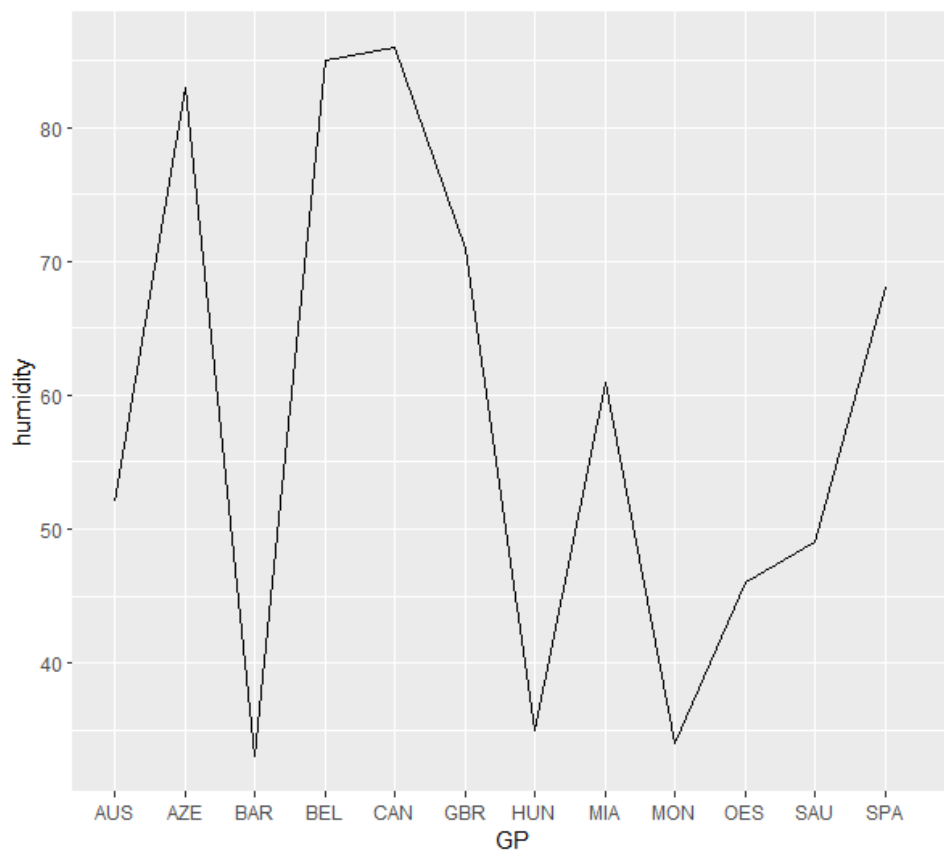


Figure 3.5: Humidity during the Qualifier of 2023

Figure 3.5 does show the humidity in percentages during the Qualifiers of 2023. With a span of the humidity from 30% to more than 90% the teams have to be prepared to act accordingly. In the case of the humidity rising over the mark of 100% the rainfall does tremendously affect the performance of the current used tyres. A switch to more adequate tyres like Intermediate or Wet tyre can improve the performance. So a tyre switch before the rainfall would be the greatest option however if the rainfall does not happen the chosen tyre will perform worse.

In the case of the Canadian Grand Prix all drivers except Verstappen drove their fastest lap using Intermediate tyres as the humidity was at more than 90%, however as the rainfall did not happen the chosen tyres by Verstappen did give him advantage while risking the rainfall happening.

Figure 3.6 does show the air pressure of the Qualifiers of the season of 2023 in the units of hPa. As the air pressure tends to stay at an even level at around 1000, for the Grand Prix of Belgium and Austria the pressure does drop down to 960 and 940 hPa. The paper of Soares and Sodre (2002) does indicate that a sinking air pressure does have significant impact on the performance of a gasoline engine. So the teams and drivers will have to be able to adapt to even the slightest changes in air pressure while not overcompensating and breaking the rules set by the organiser.

Further does the figure 3.7 present the temperature changes in between the GP's of the season of 2023 in the units of degree centigrade. The graph does show the air temperatures in orange and the track temperatures in blue. A clear connection between both graphs can be easily seen. A interesting fact however is that the air temperature is always lower than the track temperature. As the temperature does impact the performance of the cars as well as the performance of the used tyres of the drivers. With great changes in the weather ranging from 15 to almost 30 degrees in air temperature.

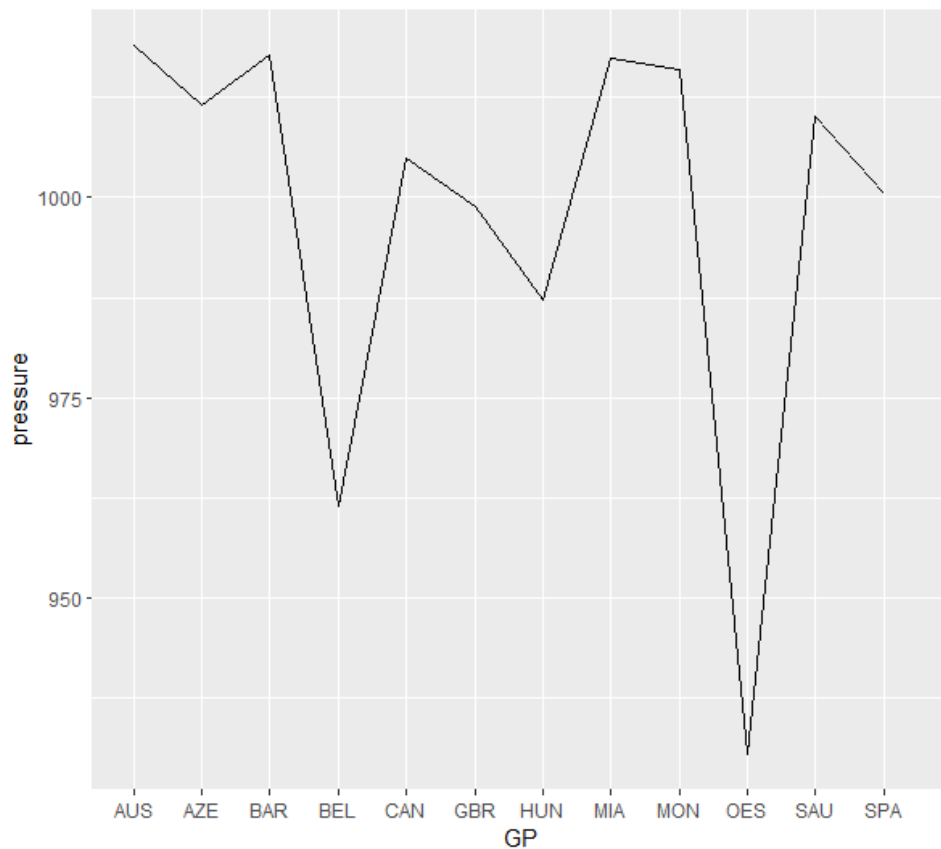


Figure 3.6: Air pressure during the Qualifier of 2023

The teams therefore have to be able to adapt to all kinds of weather situations even during the event. While the analysis of the car data have to even extend to observe external effects like weather data, it shows that the teams have to research those effects on the car and its performance. During the events the weather conditions can also switch instantaneously which have not been foreseen during the preparations of the Grand Prix. A live risk evaluation towards the tyre strategy does also affect the performance and results of the drivers.

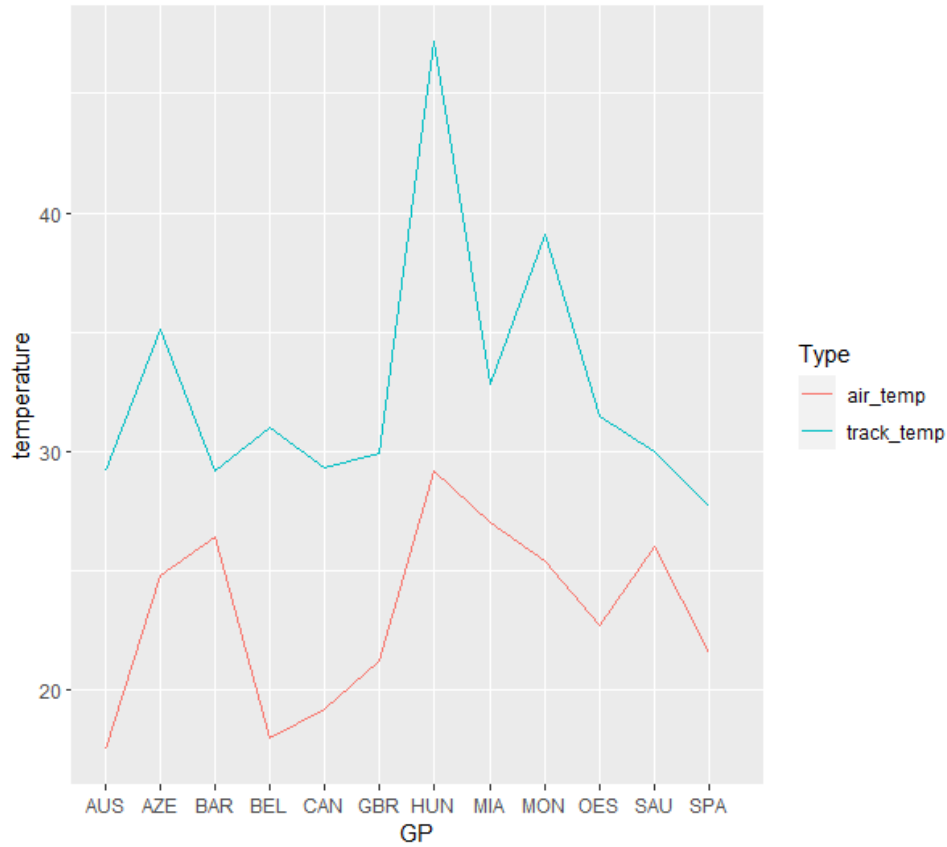


Figure 3.7: Temperatures during the Qualifier of 2023

4 Machine learning

This part of the Project work focuses on the Machine Learning model that was set up to predict the Qualifier Results of the Season of 2023. This far 12 races have been driven in the season. The usual Qualifiers are split into 3 to eliminate driver in the first 2 sessions and determine the starting positions in the last Qualifier. The data however did have some errors when showing the session of Qualifier, meaning the differentiation between Qualifier 1, 2 and 3. Because of this problem the main goal was to predict the fastest lap time of all qualifier combined for the top 4 drivers. All of the following machine learning was done in the programming language R using the machine learning framework of mlr3 created by Lang et al. (2019).

The first steps of preparing the Machine learning model were made by pre-processing the data into the right datatype and form. With this being done the model will now be able to train on a dataset of the length of more than 100,000 rows of data. As the size of data does bring its advantages it also introduces disadvantages which will be described in the following.

4.1 Learner selection

As the main target of this Machine learning model is to predict the fastest lap time of the drivers the task will be assigned to be of regression nature. After defining the task, a benchmark is set up in order to find out which learner of the mlr3 environment works best for the given data. For this procedure were the following learners were used:

- random forest learner "regr.rpart"
- regression tree learner "regr.ranger"
- linear model learner "regr.lm"
- featureless learner "regr.featureless"
- k-nearest neighbour learner "regr.kknn"

To test the learners on a further level three benchmarks were setup each using different re-sampling approaches. These methods were a Cross Validation resampling with 3 folds, a standard holdout resampling and a subsampling approach with a 4/5 split repeated 3 times. The results of the benchmarks were measured by calculating the mean squared error (mse) and root mean squared error (rmse). The results visible in figure 4.1 represent the benchmark results using the holdout resampling method. The results of this benchmark produced the best results regarding the mse value of the 3 used benchmarks. It is visible that the regression tree and linear model learner performed the best by far with the k-nearest neighbour learner coming 3rd and the random forest learner and featureless learner being 4th and 5th.

As it is not clearly visible which learner performed the best the following results were extracted using the aggregate function. The linear model learner performed best with a mse value of: 0.7970415 and rmse value of: 0.8927718. The regression tree learner did perform just slightly worse with mse and rmse values of 2.1182186 and 1.4554101.

As both of the best performing learners do also support the data type character a backup dataset was used that was not modified with the transformation of all columns into numerical or logical types. The only difference besides the datatypes are that the previously used driver number has been replaced by the driver tag.

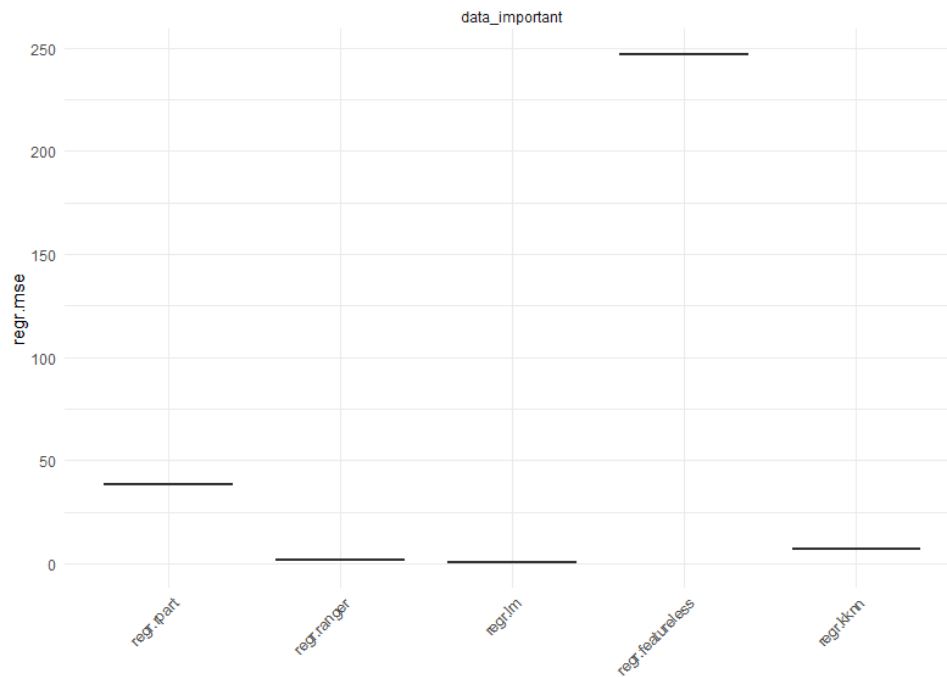


Figure 4.1: Benchmark results measured by mse

At this point another flaw in the dataset was observed, that being the feature `session_type` which differentiates between the session FP1, FP2, FP3 and R. As this information seemed useful in the first place, a prediction for the session type Q would be not manageable as it has not been present in the training data. The elimination of this feature did worsen the results of the linear model slightly by increasing the mse value to 0.8304.

4.2 Feature selection

The next step involved a feature selection process of the present dataset. The first approach was done using the importance feature of the regression tree learner which scored second best in the benchmark. This feature of the learner is automatically calculated then the learner is trained on a dataset. The importance feature ranked the sector times and speed highest followed by weather information regarding the air pressure and temperature, humidity, track temperature. The least important features were the concerned the binary variables rainfall and tyre freshness.

Coming from a logical standpoint ranking the binary variable rainfall the least important did seem counter intuitive as it stands as a important factor regarding the lap times. The variable might be partially captured by the humidity variable as rainfall is equivalent to a humidity of 100 %. As the included importance feature produced sceptical results a second method on the feature selection included a wrapper based approach using the best performing learner. The created instance included the task of the machine learning model, the linear model learner and the holdout resampling method. Mlr3 provides different feature selection optimisation methods, in the first run the sequential method was used, which reduces the features of the dataset step by step and calculating the mean squared error for each combination.

The Results of this first run did produce a new features set which eliminated temperature information completely while keeping the binary variable rainfall. This approach created a feature set with completely different significance than the extracted importance feature of the regression tree learner. The wrapper based approach with the sequential optimisation

however did produce a worse results regarding the mse value.

The next logical step would be to use a different optimisation approach like random search. As wrapper based feature selection approaches are usually very expensive regarding computing the necessary computing power for more than 100,000 rows of data did exceed the possible computational power of the used machine.

It was therefor not possible to perform the feature selection without coming across seemingly logical errors with the extraction of the importance factor or a worse performing feature set when using a computationally cheap wrapper based approach.

4.3 Results

After not being able to further improve the model with a feature selection process the prediction on the previously prepared dataset was executed. After training the the linear model learner on the dataset a prediction on the fastest lap times of each Qualifier of the season 2023 was made. The results were merged into one dataset together with the actual fastest lap times of the drivers driven during those qualifiers. The difference between both values has been calculated and plotted into the figure 4.2.

The graph does show that the difference between prediction and actual lap time does no differ more than 0.35 seconds, which does indicate a good result. It can also be observed that for most races the difference is almost always similiar between all drivers with the exception being the Canadian GP. The large difference in the Canadian GP for the prediction of Verstappen could be caused due to the chosen tyre compound, as all other driver did achieve their fastest lap time with a Intermediate tyre compound and Verstappen using a Soft compound.

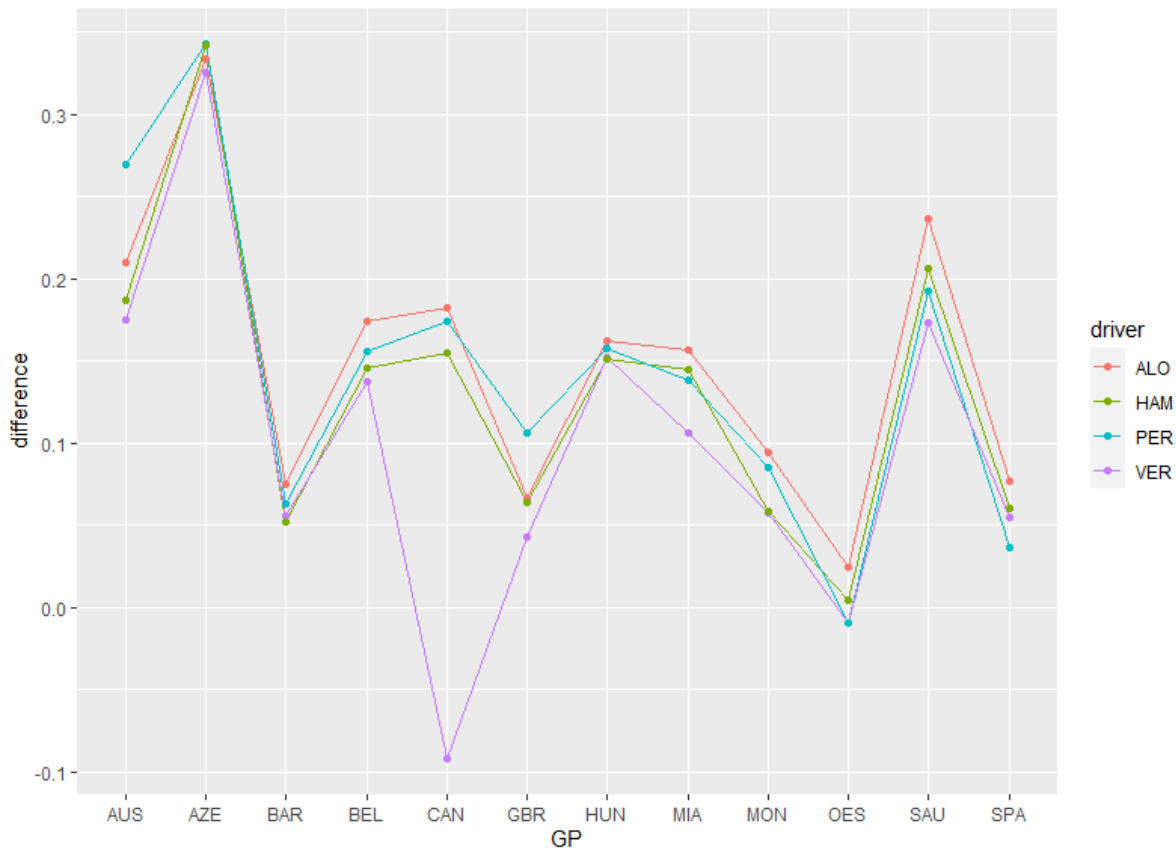


Figure 4.2: Difference between predicted and driven fastest lap time

This difference and the fact that in 2021 the Canadian GP was not included and is therefore underrepresented in terms of data could have led to the deviation of the reoccurring trend.

The largest lap time differences can be observed for the Grand Prix in Azerbaijan and Saudi Arabia. Looking into the data there can be no specific point found that could explain the great disparities from the other datapoints.

The results are showing that the difference is almost always positive meaning that the prediction is higher than actual lap time. This could be the case because the outlier detection was not calculated accurately enough leaving too many large outliers for the shorter tracks and therefore leading to a positive overshoot of the prediction.

The Australian GP shown in the figure with the tag: "OES" seemingly performed the best with generating the lowest difference of all predicted track. This is the case due to the fact that in the season of 2021 the track has been driven twice and is therefore over represented in the dataset leading to more precise data and a bias of the model towards this track.

Furthermore can be observed that the prediction of Alonso (ALO) produced the greatest differences as the orange coloured datapoints almost always seems to be above of the other drivers. This fact can again be explained with the fact that Alonso did not partake in Formula one in the season of 2021 and is therefore under represented in the dataset leading to the larger differences concerning the lap times.

Another larger disparity from the reoccurring trend that the differences in lap time are not far from one another is the datapoint of Perez in the first race in Bahrain (BAR). This could be the case due to the actual fastest lap time of Perez being almost 25% higher than the ones of the other drivers. This difference is displayed in the figure 4.3, as Perez fastest lap time was far off from the others. This spike is again breaking out of the reoccurring trend of the lap times being very close together, which could have been due to a crash or other external event. As these external events are not taken into consideration by the model it can therefore only predict the lap times under normal circumstances which leads to a larger difference in the case of an external factor changing the result of the Qualifier.

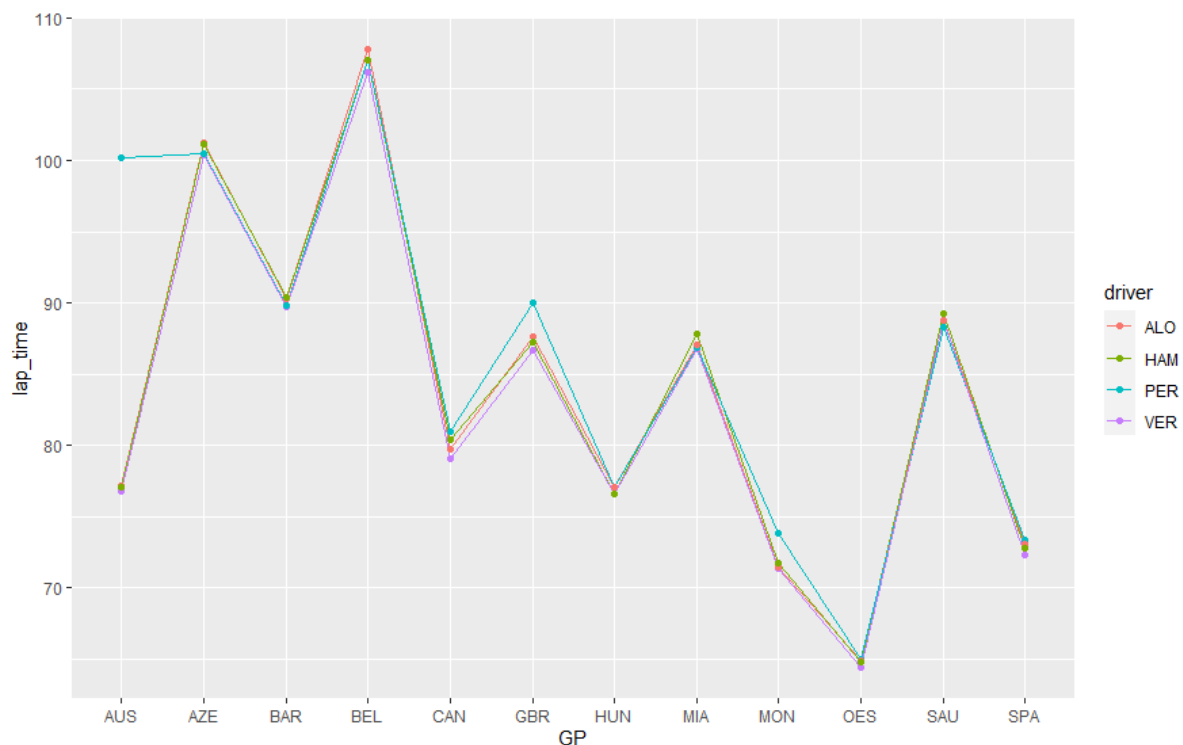


Figure 4.3: Fastest lap times of the Qualifiers of 2023

The table 4.1 shows a visual representation of the correct guesses that the machine learning model was able to make regarding the actual starting positions after the Qualifiers. The left column represents the driven GP's of 2023 while on the first row the short tags of the driver are shown that the model made predictions on. The numbers represent the order in which the Qualifiers resulted for the drivers. Meaning that if only the four drivers would be considered this would be the starting grid. For example in the Qualifier of the GP in Bahrain Verstappen secured the pole position, Perez came in second, Alonso being in the 5th place and Hamilton being last on the 7th place of the starting grid. There if only the four drivers would be considered the starting grid would look like the 2nd row of the table.

If the cells are coloured in green the machine learning model did predict the right order regarding the lap times. This process is further complicated due to the fact that the model only predicted the fastest lap of all three Qualifiers while in reality the fastest lap time of the last Qualifier decides the starting positions. Even with this additional complication the model still managed to correctly guess more than 80 % of the starting positions correctly.

Table 4.1: Comparison of correct guesses of starting grid

GP	VER	PER	ALO	HAM
BAR	1	2	3	4
SAU	4	1	2	3
AUS	1	4	3	2
AZE	1	2	3	4
MIA	3	1	2	4
MON	1	4	2	3
SPA	1	4	3	2
CAN	1	4	2	3
OES	1	4	3	2
GBR	1	4	3	2
HUN	2	4	3	1
BEL	1	2	3	4

The greatest difference can be observed for the GP of Miami where only 1 guess was made correctly. In this Qualifier Verstappen tried to secure a good lap time in the last laps of the final Qualifier as he did not perform as well in the previous laps. A crash of the driver Charles Leclerc however cancelled the last laps of the Qualifier leaving Verstappen on the 9th place of the starting grid Becker 2023. If he would have managed to finish his last laps and secure the first place the guesses would all be in the correct form. Just as in the previous part of the predictions did external events which are not considered in the model cause a disparity between the prediction and actual racing results.

5 Conclusion

The results of the constructed model show a good performance which can be partly explained by the large dataset used. Apart from special occasions with the impact of external effects, the model can predict the starting grid of these four drivers with an accuracy of more than 80%. To make good predictions, it is necessary to process every step carefully. Not only the machine learning part, but the data preparation is also important. In the beginning, we remove outliers of training data set and delete some useless variable. Since the original data set included many unnecessary information, which could worsen the prediction, the best solution is selecting only the constructive data in the model.

As the motor sport event Formula One is a high level race, it is normal that many unpredictable situations happen. External events like car crashes and weather abnormalities can impact the results of the races which the model is not able to foresee. Although it seems that it only affects the drivers slightly, the difference of lap time between the drivers results are always in a window of a second. Thus, small effect might lead to different results. However, that is why F1 is exciting and attractive for people. Everything is possible. Even though the model could guess most of the result, some parts remain unpredictable.

Our results suggests that the predicted lap times are always higher than the true fastest qualifying lap time. It is acceptable for our model since our target is to predict starting grid. The results appear that the differences are similar among four drivers, so the model can still predict the grid precisely. One reason for this is that most of training data is from practice section. In practice section, drivers would not drive every lap as fast as the qualifier run. Their goal in practice run is testing the cars. Therefore, the prediction lap time of model would be a little higher. For the further improvement, it is possible to estimate the difference between true value and prediction. The model would be able to predict the lap time of the fastest lap in qualifier run. This makes the model stronger and complete.

Throughout the machine learning part the impact of a large dataset containing more than 100,000 rows of data could be observed. On the one hand the large dataset was able to guarantee very precise predictions of the lap time. On the other hand the large dataset came with huge computational workloads when computing resampling and other optimisation algorithms. For future analysis a stronger machine with parallel computing could be installed to ensure a faster calculation process to be able to achieve better results.

Even though Formula 1 has a lot of factors that need to be considered, and the race results are very close, it is still possible to make some plausible prediction. It might seem that the differences of just 0.3 seconds tend to be a very precise prediction. However during the races just the smallest fraction of a second could make the difference of starting in the pole position or just at the second or even third place.

Another factor that has not been considered in the model are the changes of the regulations regarding the constructions of the car. For further analysis a dummy variable could be introduced indicating if the new or old regulations were in place at the time of the recorded lap.

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Work Allocation

Adrian Männel: Chapters 2, 3.4 and 4

Galib Chowdhury: introduction, data preparation, data visualisation

Ping Huang: Conclusion, part of data preparation and ML programming

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Dresden, 15th August 2023

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